

DATA INGESTION

TASK 1

OBJECTIVE

- Scenario - 1 : Load 20 records of data
- Scenario - 2 : Load 15 more new records of data
- Scenario - 3 Load 12 new records and 5 updated records from Src_File_1, Src_File_2, Src_File_3
- Total records in final table : $20 + 15 + 12 = 47$

SCENARIO - 1

Step : 1 Create postgres table (Inbound_file_data) -> final table which will hold all records

```
CREATE TABLE Inbound_file_data (  
    Record_ID SERIAL PRIMARY KEY,  
    Customer_ID INT NOT NULL,  
    Customer_Name VARCHAR(100),  
    Customer_Email VARCHAR(150),  
    Amount DECIMAL(10,2),  
  
    Orig_Source_File VARCHAR(255) NOT NULL,  
    Updt_Source_File VARCHAR(255) NOT NULL,  
    Created_On TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    Updated_On TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

Step : 2 Adding unique constraint on CUSTOMER_ID so it solves ON CONFLICT

```
ALTER TABLE Inbound_file_data  
ADD CONSTRAINT uq_customer UNIQUE (Customer_ID);
```

Step : 3 Create a temporary table called staging_inbound to collect imported data from csv and later load into final table (inbound_file_data)

```
CREATE TABLE staging_inbound (  
    Customer_ID INT,  
    Customer_Name VARCHAR(100),  
    Customer_Email VARCHAR(150),  
    Amount DECIMAL(10,2)  
);
```

Step : 4 Right click on staging_inbound and import csv from file explorer.

Step : 5 Insert the columns from staging_inbound into final table (Inbound_file_data)

```
INSERT INTO Inbound_file_data  
(Customer_ID, Customer_Name, Customer_Email, Amount, Orig_Source_File, Updt_Source_File)  
SELECT  
    Customer_ID, Customer_Name, Customer_Email, Amount,  
    'Src_File_1.csv', 'Src_File_1.csv'  
FROM staging_inbound  
ON CONFLICT (Customer_ID)  
DO UPDATE SET  
    Customer_Name = EXCLUDED.Customer_Name,  
    Customer_Email = EXCLUDED.Customer_Email,  
    Amount = EXCLUDED.Amount,  
    Updt_Source_File = EXCLUDED.Updt_Source_File,  
    Updated_On = CURRENT_TIMESTAMP;
```

Step : 6 Truncate the staging_inbound table to carry out the same data ingestion for other scenarios

```
TRUNCATE TABLE staging_inbound;
```

Result after scenario 1

> Should have 20 new records in the inbound_file_data table

Check using SQL command

```
-- check for result

SELECT COUNT(*) AS total_records FROM Inbound_file_data;

-- column wise result

SELECT Record_ID, Customer_ID, Customer_Name, Orig_Source_File, Updt_Source_File, Created_On, Updated_On
FROM Inbound_file_data
ORDER BY Customer_ID;
```

- Repeat the same steps for scenario 2 and 3

Results after 1, 2, and 3 scenarios

```
SELECT COUNT(*) AS total_records FROM Inbound_file_data;

SELECT Record_ID, Customer_ID, Customer_Name, Orig_Source_File, Updt_Source_File, Created_On, Updated_On
FROM Inbound_file_data
ORDER BY Customer_ID
LIMIT 20;
```

inbound_file_data 1

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Execution Log

SELECT record_id, Customer_id, Customer_name, orig_source_file, updt_source_file, created_on, updated_on

Enter a SQL expression to filter results (use Ctrl+Space)

Text

Record

	123 record_id	123 customer_id	A2 customer_name	A2 orig_source_file	A2 updt_source_file	created_on	updated_on
1	1		John Doe	Src_File_1.csv	Src_File_1.csv	2025-09-25 19:37:04.946	2025-09-25 19:37:04.946
2	2		Jane Smith	Src_File_1.csv	Src_File_1.csv	2025-09-25 19:37:04.946	2025-09-25 19:37:04.946
3	3		Mike Johnson	Src_File_1.csv	Src_File_1.csv	2025-09-25 19:37:04.946	2025-09-25 19:37:04.946
4	4		Susan Lee	Src_File_1.csv	Src_File_1.csv	2025-09-25 19:37:04.946	2025-09-25 19:37:04.946
5	5		David Kim	Src_File_1.csv	Src_File_1.csv	2025-09-25 19:37:04.946	2025-09-25 19:37:04.946
6	6		Emma Watson	Src_File_1.csv	Src_File_1.csv	2025-09-25 19:37:04.946	2025-09-25 19:37:04.946
7	7		Chris Evans	Src_File_1.csv	Src_File_1.csv	2025-09-25 19:37:04.946	2025-09-25 19:37:04.946
8	8		Nina Brown	Src_File_1.csv	Src_File_1.csv	2025-09-25 19:37:04.946	2025-09-25 19:37:04.946
9	9		Peter Parker	Src_File_1.csv	Src_File_1.csv	2025-09-25 19:37:04.946	2025-09-25 19:37:04.946
10	10		Tony Stark	Src_File_1.csv	Src_File_1.csv	2025-09-25 19:37:04.946	2025-09-25 19:37:04.946
11	11		Bruce Wayne	Src_File_1.csv	Src_File_1.csv	2025-09-25 19:37:04.946	2025-09-25 19:37:04.946
12	12		Clark Kent	Src_File_1.csv	Src_File_1.csv	2025-09-25 19:37:04.946	2025-09-25 19:37:04.946
13	13		Diana Prince	Src_File_1.csv	Src_File_1.csv	2025-09-25 19:37:04.946	2025-09-25 19:37:04.946
14	14		Barry Allen	Src_File_1.csv	Src_File_1.csv	2025-09-25 19:37:04.946	2025-09-25 19:37:04.946
15	15		Arthur Curry	Src_File_1.csv	Src_File_1.csv	2025-09-25 19:37:04.946	2025-09-25 19:37:04.946
16	16		Steve Rogers	Src_File_1.csv	Src_File_1.csv	2025-09-25 19:37:04.946	2025-09-25 19:37:04.946
17	17		Natasha Romanoff	Src_File_1.csv	Src_File_1.csv	2025-09-25 19:37:04.946	2025-09-25 19:37:04.946
18	18		Wanda Maximoff	Src_File_1.csv	Src_File_1.csv	2025-09-25 19:37:04.946	2025-09-25 19:37:04.946
19	19		Stephen Strange	Src_File_1.csv	Src_File_1.csv	2025-09-25 19:37:04.946	2025-09-25 19:37:04.946
20	20		Scott Lang	Src_File_1.csv	Src_File_1.csv	2025-09-25 19:37:04.946	2025-09-25 19:37:04.946

```
-- Check specific Customer_IDs
SELECT *
FROM Inbound_file_data
WHERE Customer_ID IN (101,105,110,113,118,202,209,214);
```

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